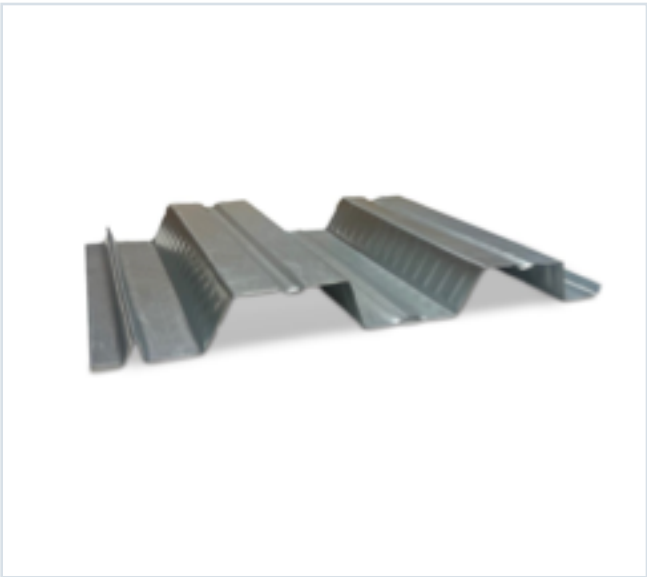


DECKING PRODUCT SUBMITTAL

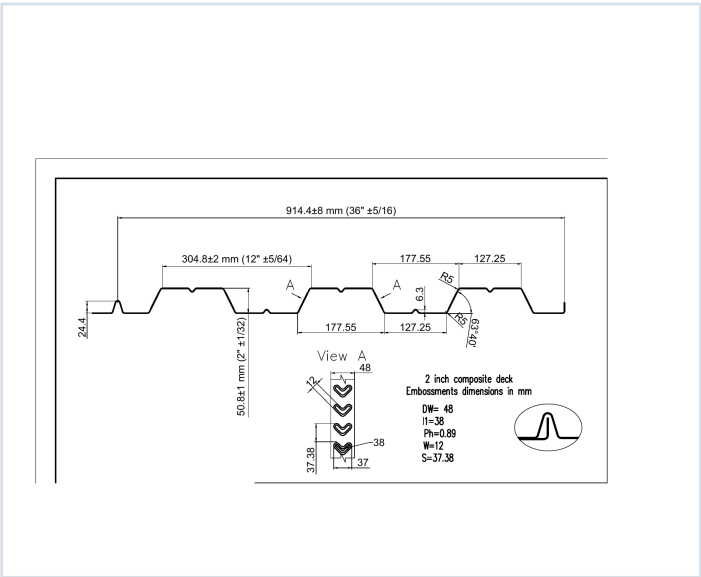
2CD-FY50-20-G40

PRODUCT	GRADE	GAUGE	COATING
2" Composite Deck	Grade 50	20 ga	G40

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 20 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0358	1.9	50	0.329	0.336	9860	10060	0.393	0.373

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 54, 55, 56, 57, 58, 59, 60, 61
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

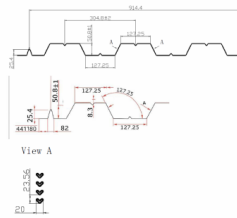
2CD-FY50-20-G40 · 20 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

STEEL CATALOG PAGE 54

GRADE 50 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (pcf)	E _s (ksi)	S _x (in ³ /ft)	S _y (in ³ /ft)	ASD (E _s + 16E _s) (in ³ /ft)	I _x (in ⁴ /ft)	I _y (in ⁴ /ft)
22	0.0295	16	50	0.247	0.254	2427	2000	0.31
20	0.0358	19	50	0.328	0.336	3880	3000	0.393
18	0.0474	25	50	0.485	0.500	14700	14000	0.548
16	0.0598	32	50	0.645	0.664	39100	39000	0.703

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/AISC 360-10, AISC 360-10C and AISC 360-10D.

Shear and Web Crippling

Gage	V _u (kips)	Web Crippling (R _v /3), kips		Web Crippling (R _v /3), kips	
		End Bearing	Interior Bearing	End Bearing	Interior Bearing
22	388	434	476	529	627
20	278	295	325	364	427
18	365	386	423	485	573
16	463	507	557	637	754

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/AISC 360-10, AISC 360-10C and AISC 360-10D.

STEEL CATALOG PAGE 55

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	137	133	127	121	115	109	103	97	91	85	79
	20	183	174	165	156	147	138	129	120	111	102	93
	18	273	258	244	230	216	202	188	174	160	146	132
	16	358	338	319	300	281	262	243	224	205	186	167
Double	22	131	125	119	113	107	101	95	89	83	77	71
	20	186	177	168	159	150	141	132	123	114	105	96
	18	277	262	248	234	220	206	192	178	164	150	136
	16	367	343	324	305	286	267	248	229	210	191	172
Triple	22	176	169	162	155	148	141	134	127	120	113	106
	20	243	233	224	215	206	197	188	179	170	161	152
	18	346	324	305	286	267	248	229	210	191	172	153
	16	446	418	390	362	334	306	278	250	222	194	166

Note: 1. All section properties and ASD (E_s + 16E_s) uniform loads are calculated in accordance with ANSI/AISC 360-10, AISC 360-10C and AISC 360-10D.
2. Loads shown in table are uniformly distributed superimposed loads in psf. Span lengths include center-to-center spacing of supports. Tabulated loads shall not be increased by assuming clear span dimensions.
3. Bending Moment formula used for flexural stress limitations are: Single and Two Span: $M = \frac{wL^2}{8}$; Three Span: $M = \frac{wL^2}{10}$.

Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

Uniform Superimposed Service Load that Causes L/240 Deflection (PSF)

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	13	13	12	11	10	9	8	7	6	5	4
	20	18	17	16	15	14	13	12	11	10	9	8
	18	27	26	24	23	21	20	19	18	16	15	14
	16	36	34	32	30	28	26	24	22	20	18	16
Double	22	20	19	18	17	16	15	14	13	12	11	10
	20	28	26	25	23	22	21	20	19	18	17	16
	18	39	37	35	33	31	29	27	25	23	21	19
	16	50	47	44	41	38	35	32	29	26	23	20
Triple	22	26	25	24	23	22	21	20	19	18	17	16
	20	36	34	32	30	28	26	24	22	20	18	16
	18	50	47	44	41	38	35	32	29	26	23	20
	16	67	63	59	55	51	47	43	39	35	31	27

Note: For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.667.

PUBLISHED PERFORMANCE DATA

2CD-FY50-20-G40 · 20 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 56

Normal Weight Concrete (145 pcf)				Lightweight Concrete (115 pcf)			
Maximum Unshored Clear Span				Maximum Unshored Clear Span			
400 (#200) 39 PSF	2x12x22 ga	7'-11"	9'-1"	9'-3"	2x12x22 ga	8'-6"	9'-9"
	2x12x20 ga	9'-3"	10'-6"	10'-10"	2x12x20 ga	10'-0"	11'-3"
	2x12x18 ga	10'-5"	12'-9"	12'-5"	2x12x18 ga	11'-2"	13'-9"
	2x12x16 ga	11'-4"	14'-2"	13'-4"	2x12x16 ga	12'-0"	14'-1"
450 (#250) 45 PSF	2x12x22 ga	7'-7"	8'-8"	8'-10"	2x12x22 ga	8'-2"	9'-5"
	2x12x20 ga	8'-10"	10'-0"	10'-4"	2x12x20 ga	9'-7"	10'-10"
	2x12x18 ga	9'-11"	12'-2"	12'-0"	2x12x18 ga	10'-9"	13'-3"
	2x12x16 ga	10'-11"	13'-8"	12'-10"	2x12x16 ga	11'-8"	14'-6"
500 (#300) 51 PSF	2x12x22 ga	7'-3"	8'-4"	8'-6"	2x12x22 ga	7'-11"	9'-1"
	2x12x20 ga	8'-6"	9'-7"	9'-11"	2x12x20 ga	9'-3"	10'-6"
	2x12x18 ga	9'-7"	11'-8"	11'-8"	2x12x18 ga	10'-5"	12'-9"
	2x12x16 ga	10'-6"	13'-3"	12'-6"	2x12x16 ga	11'-4"	14'-2"
550 (#350) 57 PSF	2x12x22 ga	6'-11"	8'-0"	8'-2"	2x12x22 ga	7'-7"	8'-9"
	2x12x20 ga	8'-3"	9'-3"	9'-6"	2x12x20 ga	8'-11"	10'-1"
	2x12x18 ga	9'-3"	11'-3"	11'-4"	2x12x18 ga	10'-0"	12'-4"
	2x12x16 ga	10'-1"	12'-9"	12'-2"	2x12x16 ga	11'-0"	13'-11"
600 (#400) 63 PSF	2x12x22 ga	6'-9"	7'-9"	7'-10"	2x12x22 ga	7'-5"	8'-6"
	2x12x20 ga	7'-11"	8'-11"	9'-2"	2x12x20 ga	8'-8"	9'-9"
	2x12x18 ga	8'-11"	10'-10"	11'-0"	2x12x18 ga	9'-9"	11'-11"
	2x12x16 ga	9'-9"	12'-4"	11'-10"	2x12x16 ga	10'-8"	13'-6"
650 (#450) 69 PSF	2x12x22 ga	6'-6"	7'-6"	7'-7"	2x12x22 ga	7'-2"	8'-3"
	2x12x20 ga	7'-9"	8'-7"	8'-10"	2x12x20 ga	8'-5"	9'-5"
	2x12x18 ga	8'-8"	10'-6"	10'-9"	2x12x18 ga	9'-5"	11'-6"
	2x12x16 ga	9'-6"	11'-11"	11'-7"	2x12x16 ga	10'-4"	12'-4"

Note
With crippling and clear span not shown accounted for in these tables. Required bearing should be determined based on specific span conditions.

STEEL CATALOG PAGE 61

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	375	330	290	260	230
4.5	35	400	400	400	400	395	377	364
5	39	400	400	400	400	400	379	357
5.5	44	400	400	400	400	400	400	393
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	210	180	172	157	143	131	120	110
4.5	246	219	205	186	174	160	147	136
5	304	275	250	227	208	191	175	161
5.5	354	320	291	265	242	222	204	188
6	400	367	334	304	278	255	234	216
6.5	400	400	377	344	314	288	265	244

16 ga Lightweight Concrete (115 pcf, F_c = 3,000 psi)

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	400	400	392	323	290
4.5	35	400	400	400	400	400	390	360
5	39	400	400	400	400	400	400	400
5.5	44	400	400	400	400	400	400	400
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	261	236	225	196	179	164	151	139
4.5	317	287	261	238	218	200	184	170
5	377	342	311	284	260	238	219	202
5.5	436	398	362	330	303	278	254	236
6	400	400	400	380	348	319	294	272
6.5	400	400	400	400	393	361	333	307